```c
#include <stdio.h>

#include <stdlib.h>

#define MAX 20   // Maximum number of vertices

// DFS function

void DFS(int adj[MAX][MAX], int visited[MAX], int start, int n) {

    visited[start] = 1;  // mark current vertex as visited

    for (int i = 0; i < n; i++) {

        if (adj[start][i] == 1 && !visited[i]) {

            DFS(adj, visited, i, n);

        }

    }

}

// Function to check connectivity

void checkConnected(int adj[MAX][MAX], int n) {

    int visited[MAX] = {0};

    // Start DFS from vertex 0

    DFS(adj, visited, 0, n);

    // Check if all vertices are visited

    int connected = 1;

    for (int i = 0; i < n; i++) {

        if (!visited[i]) {

            connected = 0;

            break;

        }

    }

    if (connected)

        printf("The graph is CONNECTED.\n");

    else

        printf("The graph is NOT CONNECTED.\n");

}

// Main function
```

```
int main() {
    int n; // number of vertices
    int adj[MAX][MAX];
    printf("Enter number of vertices: ");
    scanf("%d", &n);
    printf("Enter adjacency matrix (%d x %d):\n", n, n);
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%d", &adj[i][j]);
        }
    }
    checkConnected(adj, n);
    return 0;
}
```
OUTPUT:

```
Enter number of vertices: 4
Enter adjacency matrix (4 x 4):
0 1 0 0
1 0 0 0
0 0 0 0
0 0 1 0
The graph is NOT CONNECTED.

Process returned 0 (0x0)   execution time : 66.425 s
Press any key to continue.
```